

Chapter 23 — Event and Messaging Architecture

Field	Value
Status	Source-aligned; architecture audit passed
Version	0.1
Scope	Provide typed, secure, traceable communication for commands, queries, events, streams, and persistent collaboration across HAL identities.
Constitutional basis	Decisions 3–4, 23, 27, 32, 39–40, 44, 49
Owner review	None required

1. Purpose

Provide typed, secure, traceable communication for commands, queries, events, streams, and persistent collaboration across HAL identities.

2. Authoritative Responsibilities

Component	Authoritative responsibility
HAL Communication Protocol	Message semantics, envelopes, versioning, identity, intent, provenance, and security metadata
Event Fabric	Point-to-point, publish/subscribe, request/response, streaming, deferred delivery, and replay
Thread Service	Correlation, participants, conversation state, decisions, questions, and continuation
Delivery Controller	Priority, expiry, deduplication, retry, backpressure, and dead-letter handling

3. Core State and Records

Object	Required content
Communication Object	Sender, recipients, intent, context, priority, evidence, classification, expiry, correlation, and provenance.
Message Primitive Contract	Command, query, event, stream, or coordination semantics; ordering, persistence, acknowledgement, replay, expiry, and recovery.
Command	Requested mutation, target authority, preconditions, idempotency, deadline, and reply contract.
Event	Immutable completed fact with source identity, occurrence/receipt time, causation, and integrity.
Conversation Object	Purpose, participants, state, decisions, open questions, dependencies, confidence, and next actions.

4. Runtime Workflow

1. Authenticate sender and recipient identities and resolve the applicable authorization and Treaty boundaries.
2. Validate schema version, primitive contract, ordering/persistence semantics, intent, classification, expiry, replay protection, and routing metadata.
3. Encrypt across process, device, or trust boundaries; apply end-to-end protection for sensitive payloads.
4. Deliver by intent and capability rather than binding planners to a particular implementation.
5. Record correlation and receipt evidence; deduplicate before side-effecting command handling.
6. Persist significant collaboration in Conversation Objects independently of transcript presentation.
7. Apply bounded retry, backpressure, deferral, or dead-letter handling without losing provenance.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Unknown schema: reject or route to an explicit compatibility adapter; never guess semantics.
- Duplicate command: return the recorded disposition using its idempotency identity.
- Unavailable recipient: defer until expiry or return a transparent failure according to message policy.
- Compromised event fabric: payload encryption and signatures preserve confidentiality/integrity; revoke transport trust.

7. Constitutional Guarantees

- Communication conveys intent; evidence supports decisions; authority determines action.
- Network location never establishes trust—identity, authorization, and cryptographic proof do.
- Commands, queries, events, streams, and coordination use explicit semantics rather than one forced primitive.
- Transport infrastructure may route protected messages without automatically reading their payloads.
- Every consequential message remains attributable, correlated, versioned, and auditable.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 3–4, 23, 27, 32, 39–40, 44, 49. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.